# MOCK INTERVIEW SYSTEM

## PROJECT SYNOPSIS

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# 1. INTRODUCTION

The Mock Interview System is a web-based application designed for students and job seekers to improve their interview skills. Interviews are one of the most important stages in the recruitment process. Many candidates fail due to lack of confidence, poor communication skills, and insufficient preparation. This project provides a platform where users can practice interviews in a realistic environment.

The application allows users to select different interview levels such as Beginner, Intermediate, and Hard. The system displays interview questions and evaluates the answers. The project helps users improve confidence and technical knowledge before attending real interviews.

This system is useful for colleges, placement training centers, and fresh graduates. The platform provides flexibility for users to practice anytime and anywhere through an internet connection.

# 2. PROBLEM STATEMENT

Many students and freshers are unable to perform well during interviews because they do not get enough opportunities to practice. Traditional interview preparation methods depend on trainers and coaching centers, which are expensive and time-consuming.

Most students experience fear and nervousness during interviews. There is a need for a digital platform that can provide interview practice, performance analysis, and confidence improvement in an easy and affordable way.

# 3. OBJECTIVES OF THE PROJECT

The main objectives of the Mock Interview System are:

- To help students improve interview skills.
- To provide an online interview practice environment.
- To increase confidence during interviews.

• To reduce interview fear and nervousness.
• To evaluate user performance and scores.
• To provide different difficulty levels.
• To create a simple and user-friendly system.
• To help students prepare for placement activities.

## 4. SCOPE OF THE PROJECT

The project scope includes online interview preparation for students, graduates, and job seekers. Users can register, log in, attend mock interviews, and view their performance reports.

The system can be implemented in colleges, training institutes, and placement cells. Future versions can include AI voice interviews, video interviews, resume analysis, and real-time feedback systems.

## 5. EXISTING SYSTEM

Existing interview preparation systems mainly depend on classroom training and physical interview sessions. These systems have many disadvantages such as high cost, limited practice opportunities, and dependency on trainers.

Students do not always receive personal performance analysis, and interview practice sessions are often limited. This reduces the confidence level of candidates before real interviews.

## 6. PROPOSED SYSTEM

The proposed system is a web application that provides a digital interview practice environment. Users can attend interviews from anywhere using a computer or mobile device.

The system includes user registration, login, interview categories, performance tracking,

and score analysis. The proposed solution saves time, improves accessibility, and provides continuous practice opportunities.

## 7. KEY FEATURES

- User Registration and Login
- Multiple Interview Levels
- Real-Time Interview Questions
- Performance Evaluation
- User Dashboard
- Admin Management
- Interview Score Analysis
- User-Friendly Interface

## 8. METHODOLOGY

The project follows the Software Development Life Cycle (SDLC). The stages include requirement gathering, system design, development, testing, implementation, and maintenance.

Requirement analysis helps identify user needs. System design creates the structure of the application. Development is done using web technologies. Testing ensures that the system works correctly without errors.

## 9. SYSTEM MODULES

The main modules in the system are:

1. User Module
2. Admin Module
3. Interview Module
4. Question Management Module
5. Result Analysis Module

Each module performs specific functions to ensure smooth operation of the application.

## 10. MODULE DESCRIPTION

User Module:

Allows users to create accounts, log in, and attend interviews.

Admin Module:

Admin manages users, interview questions, and categories.

Interview Module:

Handles question display and interview process.

Question Management Module:

Stores and updates interview questions.

Result Analysis Module:

Calculates scores and generates reports.

## 11. SOFTWARE REQUIREMENTS

Operating System: Windows 10/11

Frontend: HTML, CSS, JavaScript

Backend: Node.js / PHP

Database: MySQL

Browser: Chrome or Edge

These software tools are used for building and running the web application.

## 12. HARDWARE REQUIREMENTS

Processor: Intel i3 or above

RAM: 4 GB minimum

Storage: 100 GB

Monitor: LED Display

Internet Connection: Required

The hardware requirements are minimal and affordable for most users.

## 13. TECHNOLOGIES USED

HTML is used for webpage structure. CSS is used for webpage design and styling. JavaScript provides interactivity and validation.

Node.js or PHP is used for backend processing. MySQL is used for storing user details, interview questions, and performance data.

## 14. SYSTEM ARCHITECTURE

The system architecture consists of three main components:

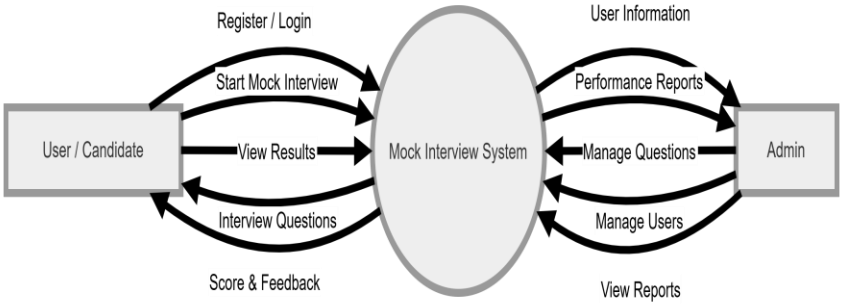
- User Interface
- Application Server
- Database Server

The user interacts with the application through the interface. Data is processed by the server and stored securely in the database.

## 15. DATA FLOW DIAGRAM (DFD)

The Data Flow Diagram represents the movement of information between users, admin, and the database.

Level 0 DFD shows the overall flow of the system.




Data Flow

**User → System**

- Register Account
- Login
- Select Interview Category
- Start Mock Interview
- Submit Answers

**System → User**

- Display Questions
- Evaluate Answers
- Generate Score
- Provide Feedback
- Show Performance Report

**Admin → System**

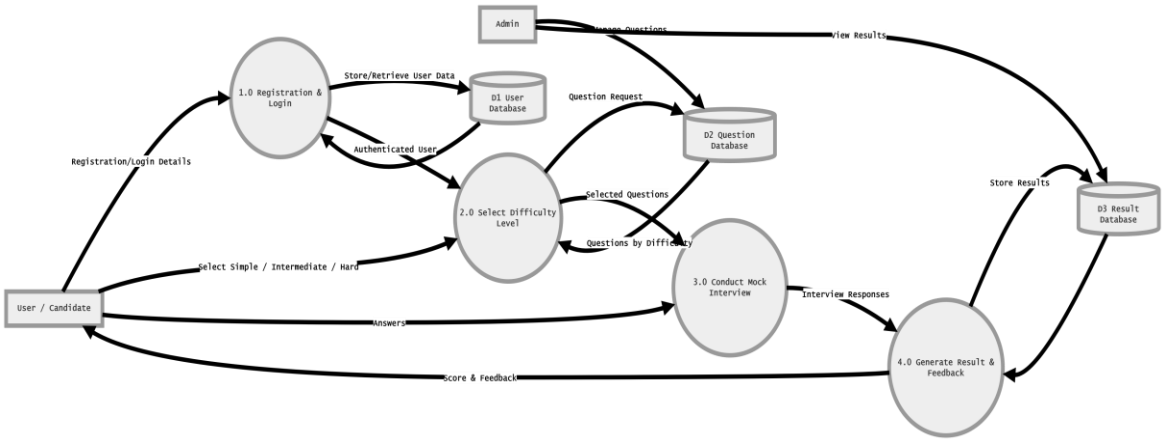
- Manage Questions
- Manage Users
- Update Interview Categories

- View Reports

**System → Admin**

- User Details
- Interview Results
- Performance Reports
- System Statistics

Level 1 DFD explains modules such as login and interviews.z




**DFD Level 1 – Process Flow (Mock Interview System)**

**Step 1: Registration & Login (1.0)**

- **User enters registration or login details.**
- **System verifies credentials using D1 - User Database.**
- **If valid, the user is allowed to proceed.**

**Step 2: Select Difficulty Level (2.0)**

- **User chooses one of the interview levels:**
    - **Simple Level**
    - **Intermediate Level**
    - **Hard Level**
- **System retrieves the corresponding questions from D2 - Question Database.**

**Step 3: Conduct Mock Interview (3.0)**

- **System displays questions based on the selected difficulty.**
- **User submits answers.**
- **System records the responses.**

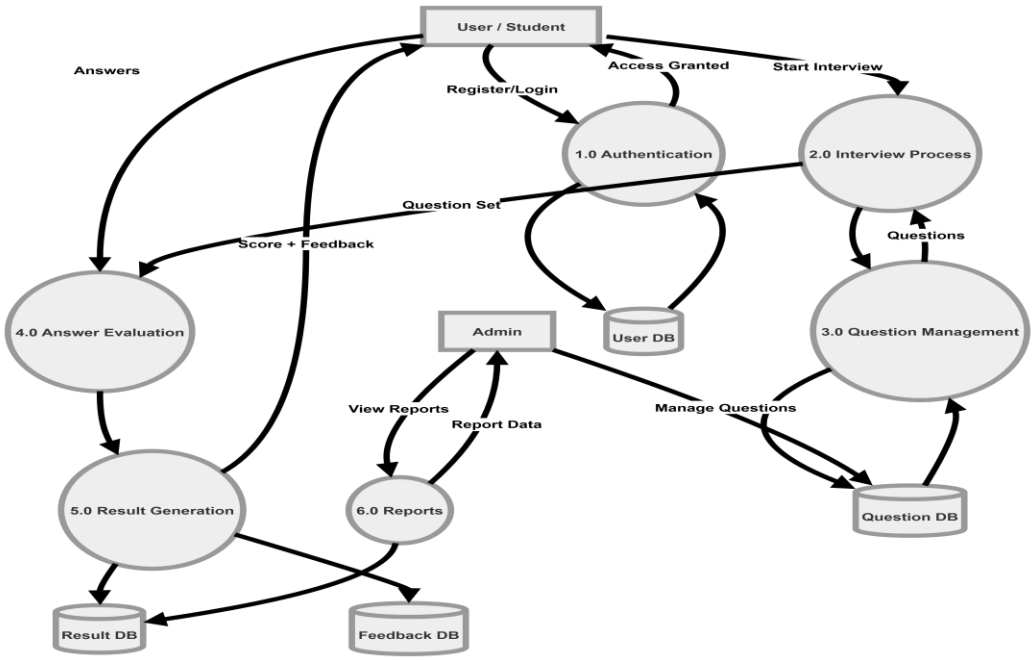
**Step 4: Generate Result & Feedback (4.0)**

- **System evaluates the answers.**
- **Score and feedback are generated.**
- **Results are stored in D3 - Result Database.**
- **User can view the final score and feedback.**

**Admin Activities**

- **Admin manages interview questions in D2 - Question Database.**
- **Admin views user performance and reports from D3 - Result Database.**

Level 2 DFD provides detailed internal processing.




**DFD Level 2 – Process Flow (Mock Interview System)**

**1.1 Register User**

- The user enters registration details such as name, email, username, and password.
- The system validates the information.
- User details are stored in **D1 – User Database**.

**1.2 Login User**

- The user enters login credentials.
- The system verifies the credentials using **D1 – User Database**.
- If valid, access to the mock interview system is granted.

**2.1 Select Difficulty**

- The user selects the interview difficulty level:
    - Simple
    - Intermediate
    - Hard
- The selected level is sent to the system.

**2.2 Retrieve Questions**

- The system retrieves questions corresponding to the selected difficulty from **D2 – Question Database**.
- Questions are prepared for the interview session.

**3.1 Attend Interview**

- The system displays interview questions one by one.
- The user reads and attempts the questions.

**3.2 Submit Answers**

- The user submits answers for all questions.
- The responses are sent to the evaluation module.

**4.1 Evaluate Answers**

- The system compares the submitted answers with the correct answers stored in **D2 – Question Database**.
- Each answer is checked and evaluated.

**4.2 Calculate Score**

- The system calculates the total score based on correct and incorrect answers.

- Performance statistics are generated.

**4.3 Generate Feedback**

- The system generates feedback based on the user's performance.
- Suggestions for improvement are prepared.

**4.4 Store Results**

- Final score and feedback are stored in **D3 – Result Database**.
- The result is displayed to the user.

**Admin Functions**

- Admin adds, updates, or deletes interview questions in **D2 – Question Database**.
- Admin views user performance reports and interview results from **D3 – Result Database**.

# 16. ER DIAGRAM

The ER Diagram represents the relationship between entities such as Users, Questions, Interviews, Results, and Admin.

It helps in understanding database structure and table relationships.

# 17. DATABASE DESIGN

The database contains multiple tables:

• User Table
• Question Table
• Result Table

• Admin Table

The database is designed to store and retrieve data efficiently and securely.

## 18. TESTING METHODS

Different testing methods are used to ensure system quality.

• Unit Testing
• Integration Testing
• System Testing
• User Acceptance Testing

Testing helps identify bugs and improves reliability of the system.

## 19. ADVANTAGES

• Easy to use
• Saves time
• Improves confidence
• Online accessibility
• Multiple difficulty levels
• Performance tracking
• User-friendly design

## 20. LIMITATIONS

• Requires internet connection
• Limited AI evaluation
• Depends on question database quality
• Basic performance analysis in current version

## 21. FUTURE ENHANCEMENTS

Future improvements include:

• AI Voice Interviews
• Video Interview Support
• Resume Analysis
• Real-Time Feedback
• Mobile Application Version
• Advanced Performance Analytics

## 22. CONCLUSION

The Mock Interview System is an effective solution for improving interview preparation. It provides a digital platform where users can practice interviews, evaluate performance, and improve confidence.

The project is useful for students and job seekers who want to prepare for real-world interviews in an efficient and affordable manner.

## 23. REFERENCES

1. HTML Documentation
2. CSS Documentation
3. JavaScript Documentation
4. Node.js Official Website
5. MySQL Documentation
6. Software Engineering Notes

**Additional Notes:**
The Mock Interview System can be expanded in future with artificial intelligence, voice recognition, and advanced analytics features to improve user experience and interview preparation quality.

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